

Haifa IPC Capital Data and Integration with Haifa Port Company (Legacy) K-Series

This report is meant to brief a new GPT/chat (and future-you) on **what the new Haifa IPC capital data is, how it relates to the existing Haifa Port Company (legacy) capital series, and how we will turn the IPC raw tables into a usable monthly K-series that can be combined with the legacy K series (and, eventually, SIPG/Bayport).**

It is both **conceptual** (what these numbers mean) and **code-oriented** (how to plug them into the existing K pipeline scripts).

1. Big-picture context

1.1 Thesis setting

- The overarching project is an **Economics Honors Thesis at UC Berkeley** on Israel's ports competition reform.
- The key empirical focus is **Haifa container capacity** around the reforms:
- The legacy public operator: **Haifa Port Company (HPC, "Haifa Port Company (legacy)" in the data).**
- The **new private entrant: Haifa Bayport**, operated by **SIPG**.
- In the background sits **Israel Ports Company (IPC)**, which owns much of the core port infrastructure (berths, yards, land, some heavy assets) and rents/leases it to operators.
- For the productivity/econometrics part of the thesis, we need a **monthly capital stock series K_t** for the Haifa container complex, so that we can compute mediators like $\log K_t$ or $\log(K_t/L_t)$ and study capital deepening alongside labor productivity.

1.2 Where we are with K for Haifa Port Company (legacy)

For **Haifa Port Company (legacy)** (HPC), we already have:

1. A **Track B PIM-based capital stock** series built from HPC's financial reports:
2. We parsed **Note 8/10 PPE tables** and cash flow notes for 2018–2024.
3. We ran them through a three-step pipeline:
 1. **Step 1 (00_build_K_step1.py)**: Standardize HPC financials, attach a capital deflator, and construct real-valued flows and stocks (K_{book_real} , I_{real} , etc.).
 2. **Step 2 (01_build_K_annual_PIM.py)**: Run a perpetual inventory model (PIM) on these annual real values to get a smoothed, depreciation-consistent annual K series (with low/central/high δ scenarios).
 3. **Step 3 (02_build_K_monthly_PIM_linear.py)**: Build a **monthly PIM-based K path** by log-linear interpolation between annual K values.

4. **Step 4 (03_build_K_monthly_PathB_lumpy.py)**: Overlay **lumpy big projects (Haifa_big_projects.csv)** on top of the smooth PIM path to create **Path B**, which explicitly adds/removes capital at project commissioning dates.

4. The output is **03_K_B_monthly_Haifa_PathB.tsv**, which contains:

5. A monthly index (2018-12-01 to 2024-12-01).
6. **K_PathB_central** (Path B monthly capital stock for HPC, in constant prices).
7. The underlying PIM baseline (**K_PIM_real_central** and lin low/central/high diagnostic columns).
8. Flags for lumpy events, gaps, imputation, etc.

Qualitatively, the HPC Path B series **declines gently** over time, reflecting:

- Moderate new investment flows.
- Relatively high depreciation.
- The 2020 sale/transfer of the marine department (tugs etc.) to IPC.

So HPC capital is **not exploding upward**; it's somewhat flat to slightly declining, even with some notable projects.

2. What the new Haifa IPC data is

We now have **new raw PPE and capex data for Haifa-related assets on IPC's books**, which we will loosely call "**Haifa IPC**" data.

2.1 Source and structure

You have extracted two tables from IPC's financial reporting (and/or an IPC subsidiary representing Haifa infrastructure):

Table 1: Net PPE by asset class

For each year 2018–2024 and several asset classes, you have:

- **year**
- **asset_class** (values like **total**, **wip**, **equip_it**, **bldg_landside**, **marine**, **land**)
- **cost_open** – opening cost balance
- **additions** – additions to cost (capex)
- **transfers** – transfers between asset classes (e.g., WIP → marine)
- **cap_interest** – capitalized borrowing costs
- **disposals** – cost of disposals
- **cost_close** – closing cost balance
- **accdep_open** – opening accumulated depreciation
- **depreciation_expense** – annual depreciation charge (sometimes split by function for some asset classes)
- **accdep_disposals** – accumulated depreciation on disposed assets

- `accdep_close` – closing accumulated depreciation
- `net_ppe_close` – closing net PPE per asset class (`cost_close` – `accdep_close`)

Crucially, the `asset_class = "total"` row gives **total net PPE** and total depreciation flows for each year.

Table 2: Depreciation and cash-flow-based investment

For the same years 2018–2024 you have a small table:

- `dep_amort_th_nis` – total depreciation and amortization.
- `purchase_ppe_cf_th_nis` – cash-flow line for purchases of PPE. This is **negative** when it is a cash outflow (like standard IFRS/GAAP cash-flow)
- `capitalized_interest_ppe_th_nis` – capitalized interest on PPE (sometimes negative, following sign conventions in the CF statement).
- `proceeds_sale_ppe_cf_th_nis` – cash inflows from sale of PPE (positive when disposing assets).
- `approx_gross_ppe_investment_th_nis` – a derived column you created that roughly reconstructs gross investment in PPE in absolute terms, using the sign-adjusted cash-flow lines.

This is an **annual, consolidated view of IPC's Haifa-related PPE** and its annual flows.

2.2 Relationship to Haifa Port Company (legacy) data

Conceptually:

- **HPC (legacy)** is the **terminal operator**, holding much of the **superstructure** on its balance sheet:
 - STS cranes and yard gantries.
 - Terminal-specific equipment.
 - Some buildings and improvements.
- **IPC** is the **infrastructure owner**:
 - Marine civil works (quays, dredging, seawalls).
 - Land, yards, some building shells.
 - Works-in-progress for large infrastructure projects.

Thus, the **HPC Path B** series we already built is only part of the story for Haifa's capital. IPC's Haifa assets represent a much larger base of **infrastructure capital** that is complementary to HPC's equipment.

Numerically, in the IPC data:

- Total net PPE (`net_ppe_close` for `asset_class = total`) is on the order of **11–13 billion NIS**.
- For HPC, total net PPE for the legacy company is on the order of **1 billion NIS**.

So when we later sum them into an “overall Haifa K”, IPC will **dominate the level** of K, while HPC will still matter for **modern equipment intensity**.

3. First-impression behavior of IPC capital

Looking at the IPC `total` net PPE data:

- **2018 → 2020:** net PPE rises (e.g., from ~11.5bn to ~12.9bn), with large WIP and marine assets.
- **2021–2023:** net PPE continues to rise but more slowly (peaking around 13.1–13.2bn).
- **2024:** net PPE dips a bit (to ~12.8bn), reflecting that depreciation plus disposals roughly match or slightly exceed net new investment.

Table 2 (approximate gross investment) shows:

- **2018–2020:** high levels of gross PPE investment (1.5bn → 0.56bn), i.e., major infrastructure push.
- **2021–2024:** lower yearly investment (roughly 0.2–0.4bn) while depreciation stays high (0.28–0.53bn), implying that **net capital formation slows** and the stock flattens or declines slightly.

This is **not inconsistent** with what we see in HPC Path B: overall Haifa capital might be **roughly stable or mildly declining** in the later years, even though there are still big projects, because much of the legacy infrastructure is ageing and depreciating.

This setup is empirically interesting:

- We can model competition and privatization in an environment where **capital deepening is not necessarily exploding**; instead, capital is reconfigured (new private terminal, reallocation of marine departments, etc.).

4. Conceptual plan: How to use IPC data in the Haifa K-series

The goal is to produce **consistent monthly K-series for IPC** that can be **combined with HPC's monthly K** to form an **overall Haifa K_t**. Later, SIPG/Bayport can be added.

4.1 Design principles

1. **Use the same deflator and price base** for IPC as for HPC.
2. HPC's Step 1 uses an OECD GFCF deflator and rebases it (e.g., to 2019 = 1.0).
3. We will apply the **same deflator** and same base-year to IPC's nominal PPE and investment flows, making all K series **comparable in real terms**.
4. **Reuse the existing PIM code as much as possible.**
5. `00_build_K_step1.py`, `01_build_K_annual_PIM.py`, `02_build_K_monthly_PIM_linear.py`, and `03_build_K_monthly_PathB_lumpy.py` are already tested for HPC.

6. They are written generically enough that we can:

- Add IPC rows to the same **master raw financials file** (`haifa_financials_raw.tsv`) and
- Extend the PIM code to **group by** `company`.

7. **Start simple for IPC: PIM-only, no IPC-specific lumpy Path B (yet).**

8. For HPC, we have detailed big-project records and we created a **lumpy Path B**.

9. For IPC, big projects are not yet fully coded at the same granularity; many are joint IPC–HPC projects or overlapping with the HPC big-project list.

10. The clean first pass is to build a **smooth PIM-based K series** for IPC only, and then consider lumpy adjustments later if needed.

11. **Combine at the end.**

12. Keep **HPC K** and **IPC K** separate in the early steps.

13. In a later script, merge the monthly series and create a **Haifa-total K series**.

5. Turning IPC raw tables into PIM-ready financials

5.1 Build a tidy IPC raw financials TSV

We want to mirror the structure expected by `00_build_K_step1.py`. That script expects a tab-separated file `haifa_financials_raw.tsv` with columns including:

- `company`
- `year`
- `ppe_net_nominal_thousands_nis_eoy`
- `depr_amort_total_thousands_nis_cf_adj`
- `purchase_of_fixed_assets_thousands_nis_cashflow`
- `proceeds_from_realization_fixed_assets_thousands_nis`
- (optionally) `purchase_of_intangible_assets_thousands_nis_cashflow`

For **IPC**, construct a new block of rows, one row per year 2018–2024:

- `company` = something like `Haifa IPC (infrastructure owner)`
- `year` = 2018, 2019, ..., 2024
- `ppe_net_nominal_thousands_nis_eoy` = `net_ppe_close` **from** `asset_class = total` in Table 1.
- `depr_amort_total_thousands_nis_cf_adj` = `dep_amort_th_nis` **from** Table 2.
- `purchase_of_fixed_assets_thousands_nis_cashflow` = `purchase_ppe_cf_th_nis` **from** Table 2 (note it is negative in the cash flow; Step 1 will re-interpret the sign as needed when defining `additions_nom`).
- `proceeds_from_realization_fixed_assets_thousands_nis` = `proceeds_sale_ppe_cf_th_nis` **from** Table 2 (positive inflows).

- `purchase_of_intangible_assets_thousands_nis_cashflow` = 0 (or populate if you later identify a clear intangible PPE line).
- `revaluation_or_basis_flag` = descriptive text like `Historical cost; IPC Haifa infrastructure (IFRS)`.
- `source_pdf`, `source_pages`, `units_note` = document the IPC reports and page references that you used.

Append these IPC rows to the existing HPC rows in `haifa_financials_raw.tsv`.

That creates a master raw financials dataset with multiple `company` values:

- `Haifa Port Company (legacy)`
- `Haifa IPC (infrastructure owner)`
- (Later: `Haifa Bayport (SIPG)` if you add it.)

5.2 Run Step 1 unchanged

`00_build_K_step1.py` is written generically:

- It reads `haifa_financials_raw.tsv`.
- It standardizes columns (renames the raw fields to `ppe_net_nom`, `additions_nom`, `depr_nom`, `disposals_nom`).
- It attaches the deflator and computes:
- `K_book_real` (real net PPE at year end).
- `I_real` (real gross investment).
- `depr_real`, `disposals_real`.

Because each row already has a `company` column, Step 1 will **process IPC rows alongside HPC rows** without special logic.

The result is a file like `00_haifa_financials_step1_real.tsv`, which now contains:

- Real-valued capital stocks and flows for **both**:
- `company = Haifa Port Company (legacy)`.
- `company = Haifa IPC (infrastructure owner)`.

6. Extending the PIM to handle IPC

6.1 Annual PIM for each company

`01_build_K_annual_PIM.py` currently:

- Reads `00_haifa_financials_step1_real.tsv`.
- Filters to HPC (`Haifa Port Company (legacy)`).
- Applies a PIM with a chosen depreciation rate δ (low/central/high) to generate an annual real K series.

- Writes `01_K_B_annual_Haifa_PIM.tsv` for HPC.

To bring IPC in, we extend this script as follows:

1. **Group by** `company`.
2. Instead of filtering to a single company, we can:
 - Either loop over unique values of `company` and run the PIM separately for each.
 - Or do a grouped PIM if that is convenient.
3. **For each company** `c`:
4. Take its annual series (`K_book_real`, `I_real`, `depr_real`, `disposals_real`) for the years 2018–2024.
5. Apply the same central δ (e.g. 6%) unless you want a different δ for infrastructure vs equipment.
6. Generate a PIM-based annual capital series:
 - `K_PIM_real_low`
 - `K_PIM_real_central`
 - `K_PIM_real_high`
7. **Write separate outputs or a stacked output:**
8. Option A: Write **one file per company** (e.g. `01_K_B_annual_Haifa_HPC_PIM.tsv` and `01_K_B_annual_Haifa_IPC_PIM.tsv`).
9. Option B: Write a **single stacked file** with columns `company`, `year`, `K_PIM_real_low`, `K_PIM_real_central`, `K_PIM_real_high`.

Given the later code, stacked with a `company` column is probably easiest, because then the monthly scripts can stay generic.

6.2 Monthly PIM path for IPC

`02_build_K_monthly_PIM_linear.py` currently:

- Reads the HPC annual PIM file.
- Creates a monthly index (2018-12-01 to 2024-12-01).
- For each year interval, it **log-linearly interpolates** between annual K values to get a smooth monthly path.

Extend it to handle multiple companies:

1. Read the **stacked annual PIM file** with `company` and `year`.
2. For each `company`:
3. Build a monthly skeleton covering its data horizon.

4. Interpolate between yearly `K_PIM_real_central` to get `K_PIM_lin_central` (and low/high if desired).
5. Concatenate results into a single monthly file, with columns:
6. `port` (Haifa)
7. `company`
8. `month` (YYYY-MM-01)
9. `K_PIM_lin_low`, `K_PIM_lin_central`, `K_PIM_lin_high`
10. Possibly diagnostic flags.

For IPC, this gives a **monthly, smooth capital stock series** in the same price base as HPC.

6.3 Path B for HPC vs PIM-only for IPC (for now)

We already have a separate script `03_build_K_monthly_PathB_lumpy.py` that:

- Reads HPC's monthly PIM skeleton.
- Reads `Haifa_big_projects.csv`.
- Builds **per-project monthly K trajectories** (`K_big.t`) based on commissioning dates and useful lives.
- Aggregates them and overlays on PIM to obtain **Path B**.

In principle we could also build a **Path B for IPC**, but that requires:

- Assigning nominal project costs between HPC and IPC for each big project.
- Knowing exactly which assets sit on IPC's books.

Since that's non-trivial, the clean first pass is:

- **Keep Path B only for HPC.**
- Use **PIM-only IPC K**.

Then, at the **combination step**, monthly Haifa-total K will be approx:

$$K_{\text{Haifa_total_central}}(t) = K_{\text{HPC_PathB_central}}(t) + K_{\text{IPC_PIM_central}}(t).$$

Later, once you have more detailed IPC project allocations, you can refine the IPC side with its own lumpy adjustments.

7. Combining HPC and IPC into an overall Haifa monthly K-series

Once the above steps are done, you will have:

- **HPC Path B monthly K:** from `03_K_B_monthly_Haifa_PathB.tsv`, with columns:
- `port = Haifa`
- `company = Haifa Port Company (legacy)`
- `month`

- `K_PathB_central`, plus PIM diagnostics.
- **IPC monthly K**: from the extended `02_build_K_monthly_PIM_linear.py`, with columns:
 - `port = Haifa`
 - `company = Haifa IPC (infrastructure owner)`
 - `month`
 - `K_PIM_lin_central` (and low/high).

7.1 New combiner script

Write a new script, e.g. `04_build_Haifa_total_K.py`, that:

1. Reads:
 2. `03_K_B_monthly_Haifa_PathB.tsv` (HPC).
 3. `02_K_B_monthly_Haifa_PIM_lin.tsv` (IPC included).
4. Filters:
5. HPC rows: `company == "Haifa Port Company (legacy)"`.
6. IPC rows: `company == "Haifa IPC (infrastructure owner)"`.
7. Outer-merge by `month` and `port`.
8. Define new variables:
9. `K_HPC_central = K_PathB_central`.
10. `K_IPC_central = K_PIM_lin_central`.
11. `K_Haifa_total_central = K_HPC_central + K_IPC_central`.

Optionally also define low/high bounds if you propagate them from both series.

1. Save an output file:
2. `04_K_Haifa_total_monthly.tsv` containing, at least:
 - `port`, `month`
 - `K_HPC_central`, `K_IPC_central`, `K_Haifa_total_central`
 - `K_HPC_low/high`, `K_IPC_low/high`, `K_Haifa_total_low/high` if you want uncertainty bands.

7.2 Using Haifa-total K with labor data

When monthly labor-hours data L_t eventually arrives, you have two options for mediators:

1. **Operator-centric K/L**
2. Use only **HPC K** as the numerator and **HPC labor** as the denominator, i.e., K_{HPC} / L_{HPC} . This measures capital per worker on the operator's books.
3. **System-level K/L**
4. Use **Haifa-total K** (HPC + IPC + later SIPG) over some system-level labor measure (e.g. all labor that actually works those assets).

For the **difference-in-differences and event-study designs**, you will likely compute:

- $\ln(K_Haifa_total_central)$ as a mediator.
- $\ln(K_Haifa_total_central / L_t)$ if you want K/L.

The key point: by aligning IPC with HPC in the same PIM framework and deflator, you get a **coherent, comparable K-series** for the Haifa port complex.

8. Future extension: adding SIPG / Haifa Bayport

Once you obtain **SIPG/Haifa Bayport** financials, you can:

1. Build a SIPG-specific raw TSV block, analogous to IPC and HPC.
2. Append SIPG rows to `haifa_financials_raw.tsv`.
3. Rerun Step 1 (real values) and the PIM scripts, now grouping by three companies.
4. For SIPG, start with PIM-only monthly K (as with IPC), unless you have SIPG-specific big-project data to build a Path B.
5. In `04_build_Haifa_total_K.py`, expand the combination to:
6.
$$K_Haifa_total_central = K_HPC_PathB_central + K_IPC_PIM_central + K_SIPG_PIM_central$$

This will give you a **fully composite K-series** for the Haifa container system, capturing public infrastructure, legacy operator superstructure, and the new private terminal.

9. Summary

- The new **Haifa IPC tables** give you annual net PPE and investment flows for the **infrastructure owner** side of Haifa.

- They are complementary to the existing **HPC (legacy) K-series**, which focuses on the operator's assets.
- By:
 - Embedding the IPC data into the existing Step 1 deflator pipeline.
 - Extending the PIM scripts to handle multiple companies.
 - Building a monthly PIM-only IPC K path.
 - Combining it with the HPC Path B monthly K.

...you will obtain a **system-level Haifa capital stock series** suitable for mediation analysis in your event-study econometric design.

This report should give any new GPT/chat enough context—both conceptual and code-level—to understand what the IPC data are, how they relate to the HPC legacy series, and how to transform them into a usable monthly K-series that can ultimately be merged with labor data and included in the thesis tables.